

Oceans form

As the young Earth slowly cooled, so did its atmosphere. Scalding steam released by volcanoes condensed to form liquid water that fell as rain, producing a downpour that lasted as long as a million years. Comets and asteroids brought yet more water. All the water pooled on the surface to form vast oceans.

▼ **WATER** *Life cannot exist without liquid water. Today water covers 71 percent of the Earth's surface.*

A watery beginning

Many scientists think life began about 3.8 billion years ago in the deep sea, which was safer than Earth's deadly surface. The first life-forms might have lived around hot volcanic vents, feeding off energy-rich chemicals dissolved in the boiling water. Special kinds of bacteria thrive in these scalding habitats even today.



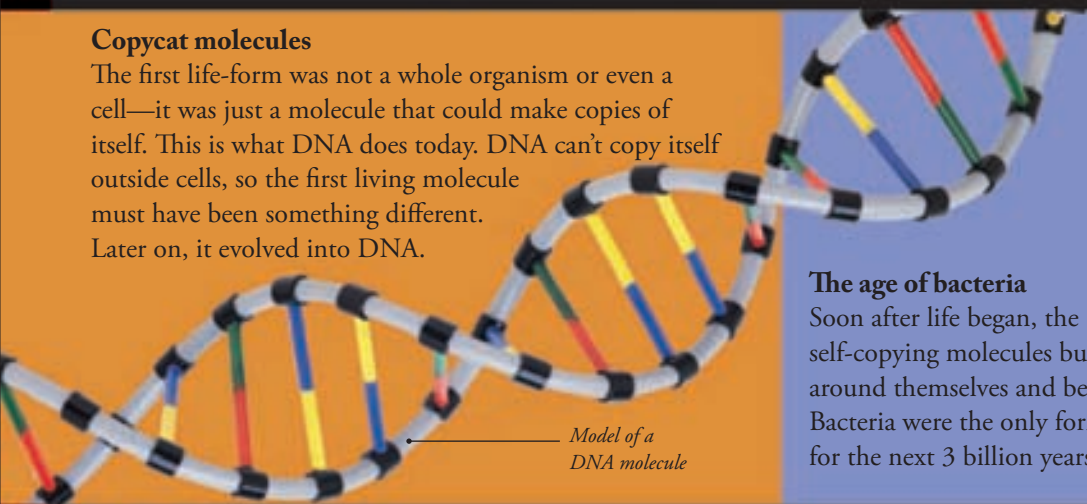
Life in hot water

At Grand Prismatic Spring in Yellowstone National Park, bacteria thrive in water too hot for any other organism to bear.



Copypat molecules

The first life-form was not a whole organism or even a cell—it was just a molecule that could make copies of itself. This is what DNA does today. DNA can't copy itself outside cells, so the first living molecule must have been something different. Later on, it evolved into DNA.



Model of a DNA molecule

Bacteria are single-celled organisms that are too small to see. Millions live on your skin and inside your body.



The age of bacteria

Soon after life began, the self-copying molecules built cells around themselves and became bacteria. Bacteria were the only forms of life on Earth for the next 3 billion years, a vast span of time.

A true survivor

Some of the oldest evidence of life on Earth comes from stromatolites. These are rocklike mounds formed by colonies of bacteria. Fossil stromatolites date back to 3.5 billion years ago. The bacteria in stromatolites live like plants, using the Sun's energy to make food and in doing so releasing oxygen. Billions of years ago, they made enough oxygen to transform Earth's air, paving the way for air-breathing animals to evolve.

▼ **LIVING STROMATOLITES** *can still be found today, such as here in Shark Bay, Western Australia.*



Stromatolite